

Dissertation on

“Adaptive Study Planning for Self-Directed Learners”

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Certificate

This is to certify that the dissertation entitled "Adaptive Study Planning for Self-Directed Learners" is a bonafide work carried out by **Rohit Saji (PES2PGE24DS201)** in partial fulfilment for the completion of Project Phase - 2 in the Program of Study – Master of Technology in Data Science and Artificial Intelligence under the rules and regulations of PES University, Bengaluru.

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Declaration

I hereby declare that the Project Phase - 2 entitled “Adaptive Study Planning for Self-Directed Learners” has been carried out under the guidance of Prof. Ramesh Prakash Guledgudd, and submitted in partial fulfilment of the course requirements for the award of the degree of Master of Technology in Data Science and Artificial Intelligence of PES University, Bengaluru, during the academic semester May – October 2026.

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Abstract

Self-directed learners routinely commit to a fixed study plan and fall behind or move ahead within days, yet the plan does not adapt and the learner has no reliable signal of how far off-track they are. Phase I of this work built an open-loop adaptive planner that estimates the learner's true study pace from session history, detects behavioural shifts in that pace, and projects the completion date with a calibrated uncertainty band, regenerating the plan when the projection diverges from the deadline. The open loop adapts to pace but never checks learning: logged hours do not prove that studying produced understanding. Phase II closes the loop with assessment-based verification of genuine learning. The system generates grounded assessments in three question families from the learner's own materials, grades each family with an authoritative arm, tracks mastery with Bayesian knowledge tracing, guides with a Socratic ladder and a gated reveal, returns advisory feedback copy, runs mixed practice through the library and practice hub, and replans from the loop verdict. Each component is evaluated against baselines and live runs, and the closed loop is validated on synthetic longitudinal histories across four learner archetypes. The selected models are realised in a working local-first web application.

Contents

1 Introduction **10**
- 1.1 Background . . . 10
- 1.2 Problem Statement . . . 10
- 1.3 Objectives . . . 11
- 1.4 Scope of the Project . . . 12
- 1.5 Organization of the Report . . . 12

2 Literature Survey **13**

3 System Requirements Specification **14**

4 Proposed Methodology **15**

5 Implementation Details **16**

6 Results and Discussion **17**

7 Conclusion and Future Work **18**

A Supplementary Tables and Figures **19**

List of Figures

List of Tables

Chapter 1

Introduction

This chapter sets out the problem this phase addresses and the shape of the work that follows. Section 1.1 recalls the Phase I system in one paragraph. Section 1.2 states the Phase II problem. Section 1.3 lists the objectives of Phase II, each with a short explanation. Section 1.4 defines the scope of the work across both phases. Section 1.5 maps the chapters that follow.

1.1 Background

A self-directed learner is a person who works through online courses, examination preparation, or self-study material without an instructor setting the pace. Such a learner typically begins with a fixed plan, of the form "two hours a day, finish by this date." Within days the plan no longer matches reality, and the fixed finish date quietly turns wrong while the plan still displays it as though it were still valid. Phase I of this project built an open-loop adaptive planner for this learner. The system estimated true study pace from session history with hierarchical Bayesian calibration, flagged genuine shifts in pace with CUSUM change-point detection, projected the completion date with a calibrated uncertainty band using Gaussian-process regression, and regenerated the schedule from realised capacity when the projection diverged from the deadline. Each component was compared against baselines on a synthetic dataset with known ground truth, using held-out evaluation with bootstrap confidence intervals and multiple-comparison correction, and the selected models were realised in a working local-first web application. The comparison reported its nulls honestly: the calibrated pace beat the fixed-rate baseline only after enough sessions had accumulated, and change detection stayed a documented robust null when no candidate dominated the frontier. What the open-loop system never checked was whether the logged hours produced learning.

1.2 Problem Statement

The pace signal in Phase I comes from self-reported session logs. It records study time and says nothing about understanding. A learner can log every planned hour and still fail every test, while the plan keeps praising the pace and the examination result collapses. Existing tools leave the same gap open from the other side. Time-tracking applications record hours without judging understanding, and static plan generators never revise against any evidence at all. The missing piece

is assessment-based verification of genuine learning, wired into the planning loop so that grades drive the decision to stay the course or replan. Phase II closes this loop. It generates grounded assessments from the learner's own materials, grades them with an authoritative arm per question family, tracks mastery from the grade history, and feeds a verdict on the plan back into schedule regeneration.

1.3 Objectives

Phase II of the project pursues the following objectives.

1. Generate grounded assessments in three families. Each generation call produces one question in a single family, objective, written, or coding, grounded in retrieved chunks of the learner's material with recorded citations. A mixed run alternates the family per call, so every question stays traceable to the material it tests.

2. Grade each family with its authoritative arm. Objective questions grade deterministically against authored answers, written answers grade against a rubric applied by a language model, and coding answers execute in a sandboxed runner with hidden tests. The server grade is authoritative. Any client-side check stays advisory and never reaches the record.

3. Estimate mastery with Bayesian knowledge tracing and steer difficulty from it. A Bayesian model updates a per-skill mastery probability from the graded response history, and the practice system adapts the difficulty of upcoming problems to the estimate instead of serving a fixed sequence.

4. Guide with a Socratic ladder and a gated reveal. The coach answers questions through a ladder of Nudge, Hint, Targeted explanation, and Reveal, and the final step stays locked until the learner confirms. Guidance therefore never hands over the answer before the learner has worked.

5. Return advisory feedback copy on each graded attempt. A server endpoint drafts learner-facing feedback grounded in the attempt and its citations for updated entries, with a static fallback when the model is unreachable. The copy never alters the grade.

6. Run mixed practice through the library and practice hub. Assessments and practice runs live under the material library with cache-first loading, and a practice run resolves its questions lazily from the loaded assessments. Starting practice therefore stays fast even on slow networks.

7. Replan from the loop verdict and validate the loop on synthetic longitudinal histories. A loop endpoint reads mastery, posture, and grade evidence and returns a hold or replan verdict under a fixed confidence gate, and a verdict for replan carries into schedule regeneration with its identity attached. Four synthetic learner archetypes over five weeks each supply the longitudinal histories that a single-semester project cannot collect from real learners.

1.4 Scope of the Project

The work is delivered across two phases. **Phase I**, reported earlier, built and evaluated the open-loop system: pace calibration, change detection, completion projection, and capacity-based schedule regeneration, each compared offline against baselines and realised in the application. **Phase II**, reported here, closes the loop with assessment-based verification of genuine learning: grounded generation, authoritative grading, mastery tracing, Socratic guidance, advisory feedback, mixed practice runs, and verdict-driven replanning, validated on synthetic longitudinal histories. The target user in both phases is the individual self-directed learner. Multi-user, institutional, and instructor-mediated settings remain out of scope, as does deployment in a real classroom. The synthetic archetypes stand in for the longitudinal real-learner study a single semester cannot supply.

1.5 Organization of the Report

The remainder of the report is organized as follows. Chapter 2 surveys the research this phase draws on: Sections 2.1 to 2.4 cover knowledge tracing, item generation and tagging, grading, and guidance with feedback, Section 2.5 notes the sourcing method, and Section 2.6 summarizes the approaches and the gaps this project addresses. Chapter 3 specifies the system requirements in Sections 3.1 to 3.4. Chapter 4 describes the proposed methodology: the closed loop in Section 4.1, the selected models in Section 4.2, the comparison method in Section 4.3, the design evolution in Section 4.4, and the system architecture in Section 4.5. Chapter 5 gives the implementation details in Sections 5.1 to 5.5. Chapter 6 presents the results and discussion in Sections 6.1 to 6.5. Chapter 7 concludes in Section 7.1 and sets out future work in Section 7.2.

Chapter 2

Literature Survey

This chapter surveys the techniques this phase draws on. Section 2.1 covers knowledge tracing and mastery models. Section 2.2 reviews automatic item generation and knowledge-component tagging. Section 2.3 describes grading methods for written and coded answers. Section 2.4 surveys Socratic guidance and learner feedback. Section 2.5 notes the method used to source the surveyed models. Section 2.6 summarizes the approaches and states the gaps this project addresses.

Chapter 3

System Requirements Specification

This chapter specifies what the system must do and what it runs on. Sections 3.1 and 3.2 list the hardware and software requirements. Sections 3.3 and 3.4 state the functional and non-functional requirements.

Chapter 4

Proposed Methodology

This chapter describes how the closed loop works and why it takes this shape. Section 4.1 presents the closed learning loop. Section 4.2 states the candidate models and the selected approach. Section 4.3 describes the comparison methodology. Section 4.4 traces the design evolution. Section 4.5 presents the system architecture.

Chapter 5

Implementation Details

This chapter gives the implementation behind the methodology. Section 5.1 records the development environment. Section 5.2 describes the datasets and their sources. Section 5.3 describes the model and service implementation. Section 5.4 states the evaluation metrics and reproducibility measures. Section 5.5 presents the Phase-2 application.

Chapter 6

Results and Discussion

This chapter presents what the evaluation found. Section 6.1 reports generation quality. Section 6.2 reports live verification. Section 6.3 reports the closed-loop decision. Section 6.4 reports synthetic longitudinal validation. Section 6.5 discusses what the numbers mean.

Chapter 7

Conclusion and Future Work

This chapter closes the report. Section 7.1 concludes what the two phases delivered. Section 7.2 sets out future work.

Appendix A

Supplementary Tables and Figures

This appendix collects the supplementary tables and figures referenced from Chapter 6. Each item repeats its in-text number so the reader can move between the discussion and the full table without searching.

Bibliography